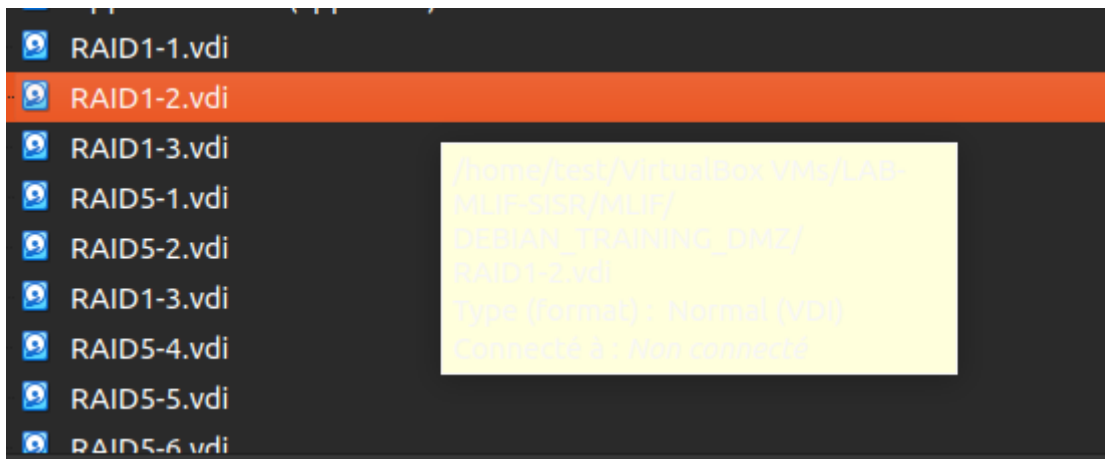


Labo RAID



disque cree

```
root@www:~# ls /dev/sd*  
/dev/sda /dev/sda1 /dev/sda2 /dev/sda5 /dev/sdb /dev/sdc /dev/sdd /dev/sde /dev/sdf /dev/sdg /dev/sdh /dev/sdi /dev/sdj  
root@www:~#
```

tout les disques presents

```
/dev/sd1 2048 2097151 2095104 1023M fd RAID Linux autodéecté  
Disque /dev/sdg : 1 GiB, 1073741824 octets, 2097152 secteurs  
Modèle de disque : VBOX HARDDISK  
Unités : secteur de 1 × 512 = 512 octets  
Taille de secteur (logique / physique) : 512 octets / 512 octets  
taille d'E/S (minimale / optimale) : 512 octets / 512 octets  
Type d'étiquette de disque : dos  
Identifiant de disque : 0x4d363072  
  
Périphérique Amorçage Début Fin Secteurs Taille Id Type  
/dev/sdg1 2048 2097151 2095104 1023M fd RAID Linux autodéecté  
  
Disque /dev/sdh : 1 GiB, 1073741824 octets, 2097152 secteurs  
Modèle de disque : VBOX HARDDISK  
Unités : secteur de 1 × 512 = 512 octets  
Taille de secteur (logique / physique) : 512 octets / 512 octets  
taille d'E/S (minimale / optimale) : 512 octets / 512 octets  
Type d'étiquette de disque : dos  
Identifiant de disque : 0xda0bbf44  
  
Périphérique Amorçage Début Fin Secteurs Taille Id Type  
/dev/sdh1 2048 2097151 2095104 1023M fd RAID Linux autodéecté  
  
Disque /dev/sdi : 1 GiB, 1073741824 octets, 2097152 secteurs  
Modèle de disque : VBOX HARDDISK  
Unités : secteur de 1 × 512 = 512 octets  
Taille de secteur (logique / physique) : 512 octets / 512 octets  
taille d'E/S (minimale / optimale) : 512 octets / 512 octets  
Type d'étiquette de disque : dos  
Identifiant de disque : 0x4b1ac5f3  
  
Périphérique Amorçage Début Fin Secteurs Taille Id Type  
/dev/sdi1 2048 2097151 2095104 1023M fd RAID Linux autodéecté  
  
Disque /dev/sdj : 1 GiB, 1073741824 octets, 2097152 secteurs  
Modèle de disque : VBOX HARDDISK  
Unités : secteur de 1 × 512 = 512 octets  
Taille de secteur (logique / physique) : 512 octets / 512 octets  
taille d'E/S (minimale / optimale) : 512 octets / 512 octets  
Type d'étiquette de disque : dos  
Identifiant de disque : 0x39ec28c2
```

partition créée + RAID LINUX AUTO DETECT

STOP2:

```
modprobe raid1  
root@www:~# modprobe raid1  
root@www:~# _
```

modprobe raid1

```

root@www:~# mdadm --create /dev/md0 --level=1 --raid-devices=2 --spare-device=1 --assume-clean /dev/sdb1 /dev/sdc1 /dev/sdd1
mdadm: Note: this array has metadata at the start and
may not be suitable as a boot device.  If you plan to
store '/boot' on this device please ensure that
your boot-loader understands md/v1.x metadata, or use
--metadata=0.90
mdadm: largest drive (/dev/sdd1) exceeds size (1046528K) by more than 1%
Continue creating array?
Continue creating array? (y/n) y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
root@www:~# _

```

```

root@www:~# mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
  Creation Time : Fri Jan 23 16:13:44 2026
    Raid Level : raid1
    Array Size : 1046528 (1022.00 MiB 1071.64 MB)
  Used Dev Size : 1046528 (1022.00 MiB 1071.64 MB)
    Raid Devices : 2
  Total Devices : 3
  Persistence : Superblock is persistent

    Update Time : Fri Jan 23 16:13:44 2026
      State : clean
  Active Devices : 2
 Working Devices : 3
 Failed Devices : 0
  Spare Devices : 1

Consistency Policy : resync

    Name : www:0 (local to host www)
    UUID : 1cf8fccf:ceed31f1:da0383d9:fc7029a1
    Events : 0

   Number   Major   Minor   RaidDevice State
    0         8       17         0     active sync   /dev/sdb1
    1         8       33         1     active sync   /dev/sdc1
    2         8       49         -     spare        /dev/sdd1
root@www:~#

```

```

root@www:~# mkfs.ext4 /dev/md0
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 261632 4k blocks and 65408 inodes
Filesystem UUID: 2bbf2906-c7fc-4d69-b860-9573ef07704c
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

```

```

root@www:~# mkdir /raid1/
root@www:~# mount /dev/md0 /raid1/
root@www:~# _

```

```
bash: /raid1/: Aucun fichier ou dossier de ce type
root@www:~# ls /raid1/
lost+found
root@www:~#
```

```
root@www:~# mount /dev/md0 /raid1/
root@www:~# df -h | grep md0
/dev/md0          988M      24K   921M    1% /raid1
root@www:~# _
```

```
root@www:~# cp -R /usr/bin/* /raid1/
root@www:~# _
```

```
root@www:~# mdadm /dev/md0 --fail /dev/sdb1
[ 1573.080079] md/raid1:md0: Disk failure on sdb1, disabling device.
[ 1573.080079] md/raid1:md0: Operation continuing on 1 devices.
mdadm: set /dev/sdb1 faulty in /dev/md0
root@www:~#
```

```
root@www:~# mdadm --detail /dev/md0
/dev/md0:
```

```
    Version : 1.2
  Creation Time : Fri Jan 23 16:13:44 2026
    Raid Level : raid1
    Array Size : 1046528 (1022.00 MiB 1071.64 MB)
  Used Dev Size : 1046528 (1022.00 MiB 1071.64 MB)
    Raid Devices : 2
  Total Devices : 3
  Persistence : Superblock is persistent

    Update Time : Fri Jan 23 16:27:52 2026
      State : clean
  Active Devices : 2
 Working Devices : 2
  Failed Devices : 1
   Spare Devices : 0
```

Consistency Policy : resync

```
    Name : www:0 (local to host www)
   UUID : 1cf8fccf:ceed31f1:da0383d9:fc7029a1
  Events : 21
```

Number	Major	Minor	RaidDevice	State	
2	8	49	0	active sync	/dev/sdd1
1	8	33	1	active sync	/dev/sdc1
0	8	17	-	faulty	/dev/sdb1

```
root@www:~#
```

```
root@www:~# cat /proc/mdstat
Personalities : [linear] [multipath] [raid0] [raid1] [raid6] [raid5] [raid4] [raid10]
md0 : active raid1 sdd1[2] sdc1[1] sdb1[0](F)
      1046528 blocks super 1.2 [2/2] [UU]

unused devices: <none>
root@www:~#
```

```
root@www:~# mdadm /dev/md0 --fail /dev/sdc1
[ 1857.997754] md/raid1:md0: Disk failure on sdc1, disabling device.
[ 1857.997754] md/raid1:md0: Operation continuing on 1 devices.
mdadm: set /dev/sdc1 faulty in /dev/md0
root@www:~# _
```

```
root@www:~# mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
  Creation Time : Fri Jan 23 16:13:44 2026
    Raid Level : raid1
    Array Size : 1046528 (1022.00 MiB 1071.64 MB)
  Used Dev Size : 1046528 (1022.00 MiB 1071.64 MB)
    Raid Devices : 2
  Total Devices : 3
  Persistence : Superblock is persistent

    Update Time : Fri Jan 23 16:32:32 2026
      State : clean, degraded
  Active Devices : 1
 Working Devices : 1
 Failed Devices : 2
  Spare Devices : 0

Consistency Policy : resync

    Name : www:0 (local to host www)
   UUID : 1cf8fccf:ceed31f1:da0383d9:fc7029a1
  Events : 23

   Number   Major   Minor   RaidDevice State
     2         8        49         0   active sync  /dev/sdd1
     -         0         0         1   removed
     0         8        17         -   faulty    /dev/sdb1
     1         8        33         -   faulty    /dev/sdc1
root@www:~#
```

```
root@www:~# mdadm /dev/md0 --remove /dev/sdb1
mdadm: hot removed /dev/sdb1 from /dev/md0
root@www:~# mdadm /dev/md0 --remove /dev/sdc1
mdadm: hot removed /dev/sdc1 from /dev/md0
root@www:~# _
```

```
root@www:~# mdadm /dev/md0 --add /dev/sdc1
mdadm: added /dev/sdc1
root@www:~# mdadm /dev/md0 --add /dev/sdb1
mdadm: added /dev/sdb1
root@www:~#
```

```

root@www:~# mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
  Creation Time : Fri Jan 23 16:13:44 2026
    Raid Level : raid1
    Array Size : 1046528 (1022.00 MiB 1071.64 MB)
  Used Dev Size : 1046528 (1022.00 MiB 1071.64 MB)
    Raid Devices : 2
  Total Devices : 3
 Persistence : Superblock is persistent

 Update Time : Fri Jan 23 16:35:26 2026
   State : clean
 Active Devices : 2
Working Devices : 3
 Failed Devices : 0
 Spare Devices : 1

Consistency Policy : resync

           Name : www:0 (local to host www)
          UUID : 1cf8fccf:ceed31f1:da0383d9:fc7029a1
        Events : 47

   Number   Major   Minor   RaidDevice State
     2         8       49           0  active sync  /dev/sdd1
     3         8       33           1  active sync  /dev/sdc1
     4         8       17           -   spare      /dev/sdb1
root@www:~# _

```

uuid:

```

UUID="2bbf2906-c7fc-4d69-b860-9573ef07704c" /raid1/      ext4      defaults 0 1

root@www:~# cat /proc/mdstat
Personalities : [raid1] [linear] [multipath] [raid0] [raid6] [raid5] [raid4] [raid10]
md127 : active raid1 sdb1[4](S) sdd1[2] sdc1[3]
      1046528 blocks super 1.2 [2/2] [UU]

```

Stop 3:

```

root@www:~# mdadm --create /dev/md1 --level=5 --raid-devices=5 --spare-devices=1 --assume-clean /dev/sde1 /dev/sdf1 /dev/sdg1 /dev/sdh1 /dev/sdi1 /dev/sdj1
mdadm: largest drive (/dev/sde1) exceeds size (1030144K) by more than 1%
Continue creating array? y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md1 started.
root@www:~# _

```

```

root@www:~# mdadm --detail /dev/md1
/dev/md1:
    Version : 1.2
    Creation Time : Fri Jan 23 17:15:42 2026
    Raid Level : raid5
    Array Size : 4120576 (3.93 GiB 4.22 GB)
    Used Dev Size : 1030144 (1006.00 MiB 1054.87 MB)
    Raid Devices : 5
    Total Devices : 6
    Persistence : Superblock is persistent

    Update Time : Fri Jan 23 17:15:42 2026
    State : clean
    Active Devices : 5
    Working Devices : 6
    Failed Devices : 0
    Spare Devices : 1


    Layout : left-symmetric
    Chunk Size : 512K

Consistency Policy : resync

    Name : www:1 (local to host www)
    UUID : 554748ab:00988b66:60ff6de3:97367b3f
    Events : 0

```

Number	Major	Minor	RaidDevice	State	
0	8	65	0	active sync	/dev/sde1
1	8	81	1	active sync	/dev/sdf1
2	8	97	2	active sync	/dev/sdg1
3	8	113	3	active sync	/dev/sdh1
4	8	129	4	active sync	/dev/sdi1
5	8	145	-	spare	/dev/sdj1

```

root@www:~# _

```

```

root@www:~# mkfs.ext4 /dev/md1
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 1030144 4k blocks and 257536 inodes
Filesystem UUID: b2c724b2-6a0f-4b4b-84c7-86fcc6beb5f9
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

```

```

root@www:~# mkdir raid5

```

```

root@www:~# _

```

```

root@www:~# mount /dev/md1 /raid5/

```

```
root@www:~# df -h /raid5/
Sys. de fichiers Taille Utilisé Dispo Uti% Monté sur
/dev/md1          3,8G      24K   3,6G   1% /raid5
root@www:~#
```

```
root@www:~# cp -R /usr/bin/* /raid5/
root@www:~#
```

```
root@www:~# mdadm /dev/md1 --fail /dev/sde1
[ 2281.381273] md/raid:md1: Disk failure on sde1, disabling device.
[ 2281.381520] md/raid:md1: Operation continuing on 4 devices.
mdadm: set /dev/sde1 faulty in /dev/md1
root@www:~#
```

```
mdadm: set /dev/sde1 faulty in /dev/md1
root@www:~# mdadm --detail /dev/md1
/dev/md1:
    Version : 1.2
  Creation Time : Fri Jan 23 17:15:42 2026
    Raid Level : raid5
    Array Size : 4120576 (3.93 GiB 4.22 GB)
  Used Dev Size : 1030144 (1006.00 MiB 1054.87 MB)
    Raid Devices : 5
  Total Devices : 6
 Persistence : Superblock is persistent

    Update Time : Fri Jan 23 17:24:28 2026
      State : clean
    Active Devices : 5
  Working Devices : 5
  Failed Devices : 1
   Spare Devices : 0


    Layout : left-symmetric
   Chunk Size : 512K
```

Consistency Policy : resync

```
    Name : www:1 (local to host www)
   UUID : 554748ab:00988b66:60ff6de3:97367b3f
  Events : 21
```

Number	Major	Minor	RaidDevice	State	
5	8	145	0	active sync	/dev/sdj1
1	8	81	1	active sync	/dev/sdf1
2	8	97	2	active sync	/dev/sdg1
3	8	113	3	active sync	/dev/sdh1
4	8	129	4	active sync	/dev/sdi1
0	8	65	-	faulty	/dev/sde1

```
root@www:~# _
```

```
root@www:~# mdadm /dev/md1 --fail /dev/sdi1
[ 2342.700357] md/raid:md1: Disk failure on sdi1, disabling device.
[ 2342.700614] md/raid:md1: Operation continuing on 4 devices.
mdadm: set /dev/sdi1 faulty in /dev/md1
root@www:~#
```

```

root@www:~# mdadm --detail /dev/md1
/dev/md1:
    Version : 1.2
    Creation Time : Fri Jan 23 17:15:42 2026
    Raid Level : raid5
    Array Size : 4120576 (3.93 GiB 4.22 GB)
    Used Dev Size : 1030144 (1006.00 MiB 1054.87 MB)
    Raid Devices : 5
    Total Devices : 6
    Persistence : Superblock is persistent

    Update Time : Fri Jan 23 17:25:19 2026
    State : clean, degraded
    Active Devices : 4
    Working Devices : 4
    Failed Devices : 2
    Spare Devices : 0


    Layout : left-symmetric
    Chunk Size : 512K

Consistency Policy : resync

    Name : www:1 (local to host www)
    UUID : 554748ab:00988b66:60ff6de3:97367b3f
    Events : 23

   Number   Major   Minor   RaidDevice State
     5         8       145         0   active sync   /dev/sdj1
     1         8        81         1   active sync   /dev/sdf1
     2         8        97         2   active sync   /dev/sdg1
     3         8       113         3   active sync   /dev/sdh1
     -         0         0         4   removed

     0         8        65         -   faulty   /dev/sde1
     4         8       129         -   faulty   /dev/sdi1

```

```

root@www:~# mdadm /dev/md1 --remove /dev/sdi1
mdadm: hot removed /dev/sdi1 from /dev/md1
root@www:~# mdadm /dev/md1 --remove /dev/sde1
mdadm: hot removed /dev/sde1 from /dev/md1

```

```

root@www:~# mdadm /dev/md1 --add /dev/sde1
mdadm: added /dev/sde1
root@www:~# mdadm /dev/md1 --add /dev/sdi1
mdadm: added /dev/sdi1

```

```

: UUID="b2c724b2-6a0f-4b4b-84c7-86fcc6beb5f9"

```